EXPERIMENT 6

1. Write a function that takes a number and returns it's reverse

```
<!DOCTYPE html>

<html>

<head>

<title>Reverse Number</title>

</head>

<body>

<script>

function reverseNumber(num) {

let reverse = 0;

while (num > 0) {

let digit = num % 10;

reverse = reverse * 10 + digit;

num = Math.floor(num / 10);

}

return reverse;

}

let result = reverseNumber(1234);

document.write("Reversed number is: " + result);

</script>

</body>

</html>
```

2. **Write function that removes duplicate elements from an array using loops**

```
<!DOCTYPE html>

<html>

<head>

<title>Remove Duplicates</title>

</head>

<body>


<h2>Remove Duplicate Elements</h2>


<p id="output"></p>


<button onclick="removeDuplicates()">Remove Duplicates</button>


<script>

function removeDuplicates() {

let arr = [1, 2, 3, 2, 4, 1, 5];

let unique = [];


for (let i = 0; i < arr.length; i++) {

let isDuplicate = false;
```

```
for (let j = 0; j < unique.length; j++) {

if (arr[i] === unique[j]) {

isDuplicate = true;

break;

}

}


if (!isDuplicate) {

unique.push(arr[i]);

}

}


document.getElementById("output").innerText =

"Original Array: " + arr + "\n" +

"Array without duplicates: " + unique;

}

</script>


</body>

</html>
```

3. Write a function to check if two strings are anagrams

```
<!DOCTYPE html>

<html>

<head>

<title>Anagram Checker</title>

</head>

<body>

<h2>Check Anagram</h2>


<label>Enter First String:</label>

<input type="text" id="str1"><br><br>


<label>Enter Second String:</label>

<input type="text" id="str2"><br><br>


<button onclick="checkAnagram()">Check</button>


<p id="result"></p>


<script>

function checkAnagram() {


let str1 = document.getElementById("str1").value;

let str2 = document.getElementById("str2").value;


// Convert to lowercase and remove spaces

str1 = str1.toLowerCase().replace(/\s/g, "");
```

```
str2 = str2.toLowerCase().replace(/\s/g, "");

if (str1.length !== str2.length) {
document.getElementById("result").innerText =
"Not Anagrams";
return;
}

let sorted1 = str1.split("").sort().join("");
let sorted2 = str2.split("").sort().join("");

if (sorted1 === sorted2) {
document.getElementById("result").innerText =
"They are Anagrams ";
} else {
document.getElementById("result").innerText =
"They are NOT Anagrams ";
}
}
</script>
</body>
</html>
```

4. Create your own version of array.map() using loops

```
<!DOCTYPE html>
<html>
<head>
<title>Custom Map Function</title>
</head>
<body>

<h2>Custom Version of array.map()</h2>

<p id="output"></p>

<button onclick="runCustomMap()">Run Custom Map</button>

<script>
// Custom map function
function myMap(arr, callback) {
let result = [];

for (let i = 0; i < arr.length; i++) {
result.push(callback(arr[i], i, arr));
}

return result;
}

function runCustomMap() {
let numbers = [1, 2, 3, 4, 5];

// Example: multiply each number by 2
let newArray = myMap(numbers, function(value) {
```

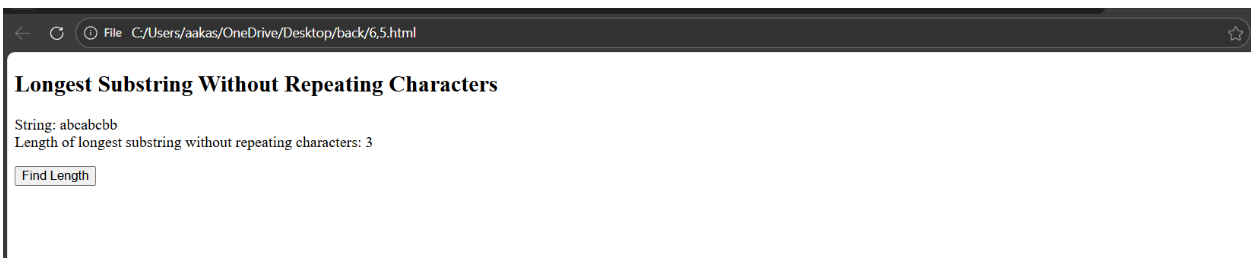
```
return value * 2;
});

document.getElementById("output").innerText =
"Original Array: " + numbers + "\n" +
"After Custom Map (x2): " + newArray;
}
</script>

</body>
</html>
```

5. Write function to find length of longest substring without repeating characters

```
<!DOCTYPE html>

<html>

<head>

<title>Longest Substring Without Repeating Characters</title>

</head>

<body>


<h2>Longest Substring Without Repeating Characters</h2>


<p id="result"></p>


<button onclick="findLongest()">Find Length</button>
```

```
<script>
function longestSubstringLength(str) {
let maxLength = 0;
let currentSubstring = "";

for (let i = 0; i < str.length; i++) {

let index = currentSubstring.indexOf(str[i]);

if (index !== -1) {
currentSubstring = currentSubstring.slice(index + 1);
}

currentSubstring += str[i];

if (currentSubstring.length > maxLength) {
maxLength = currentSubstring.length;
}
}

return maxLength;
}

function findLongest() {
let text = "abcabcbb";
```

```
let length = longestSubstringLength(text);


document.getElementById("result").innerText =

"String: " + text + "\n" +

"Length of longest substring without repeating characters: " + length;

}

</script>


</body>

</html>
```



6. Write a function that flatens nested array of any depth using recursion

```
<!DOCTYPE html>

<html>

<head>

<title>Flatten Nested Array</title>

</head>

<body>


<h2>Flatten Nested Array Using Recursion</h2>


<p id="result"></p>
```

```
<button onclick="runFlatten()">Flatten Array</button>

<script>
// Recursive function
function flattenArray(arr) {
let result = [];

for (let i = 0; i < arr.length; i++) {

if (Array.isArray(arr[i])) {
// If element is array, call function again
let flatPart = flattenArray(arr[i]);

for (let j = 0; j < flatPart.length; j++) {
result.push(flatPart[j]);
}

} else {
result.push(arr[i]);
}
}

return result;
}
```

```
function runFlatten() {

let nestedArray = [1, [2, [3, 4], 5], [6, [7, [8]]]];


let flatArray = flattenArray(nestedArray);


document.getElementById("result").innerText =

"Original Array: " + JSON.stringify(nestedArray) + "\n" +

"Flattened Array: " + flatArray;

}

</script>


</body>

</html>
```

7. Create your own debounce function

```
<!DOCTYPE html>

<html>

<head>

<title>Custom Debounce Function</title>

</head>

<body>


<h2>Debounce Example</h2>
```

```
}

<input type="text" id="search" placeholder="Type something...">
<p id="result"></p>

<script>
// Custom debounce function
function debounce(func, delay) {
let timer;

return function() {
clearTimeout(timer);

timer = setTimeout(() => {
func();
}, delay);
};
}

// Function to run after delay
function showMessage() {

let value = document.getElementById("search").value;

document.getElementById("result").innerText =
"You typed: " + value;
}
```

```
// Create debounced version (1 second delay)

let debouncedFunction = debounce(showMessage, 1000);


// Add event listener

document.getElementById("search")

.addEventListener("keyup", debouncedFunction);


</script>


</body>

</html>
```